WHAT IF?! EP 05

FREE REPORT CARD
you commented TOOT

The Gas Report Card



Your gut, by the numbers. What is in a fart, what the one real study measured, and five myths busted. Graded by me, sourced by science.

WHAT IS IN A FART

Nitrogen + CO2
big chunks of the mix. They do not burn

Hydrogen
the biggest gas the study measured. It burns

Methane
burns, but only about a third of people make it

Sulphur gases
a tiny trace, and the whole smell

REPORT CARD: ONE HUMAN, ONE DAY

Subject	Grade	Teacher's note
Volume	B	Median **90 mL** per fart (range 33 to 125 mL) in the 1991 Tomlin study of 10 volunteers. About 705 mL over 24 hours (range 476 to 1,491).
Frequency	C+	Median **8** daytime episodes in that study. The volunteers also ate 200 g of baked beans.
Flammability	A	Hydrogen (median 361 mL a day) and methane burn. Bowel gas has even ignited during surgery and burned a patient.
Smell	F	A tiny trace of sulphur gas does all of it. Fire and smell are different departments.
Planet impact	incomplete	Everyone at once is about **295 Olympic pools** of gas. Earth would not notice. The real problem is cows: livestock is about 12 to 14.5% of human greenhouse gas, mostly burped.
History	A+	**Le Pétomane** (Joseph Pujol, 1857 to 1945) headlined the Moulin Rouge in the 1890s. **Benjamin Franklin** wrote a 1781 essay asking scientists to make farts smell nice.

HISTORY CORNER



Le Pétomane
Moulin Rouge star of the 1890s. Reports call him one of the best-paid entertainers of his day.



Ben Franklin
1781 satire to a Brussels academy. He never sent it there; it went to Richard Price in 1783.



Emperor Claudius
Reportedly planned an edict allowing farts at table. A historian thinks he was joking.

Sources Tomlin et al., Gut 1991 (pubmed.ncbi.nlm.nih.gov/1648028) · UN World Population Prospects 2024 (8.2 bn) · Wikipedia: Le Pétomane; A Letter to a Royal Academy · FAO 2013 via Phys.org (livestock emissions) · Gizmodo AU; Medicine Today 2005



Five fart myths, busted

The ones that keep going round, and what the sources actually say.

MYTH 1 **"Smelling farts is good for you."** A 2014 Exeter study tested a compound that releases hydrogen sulphide, and it protected mitochondria in lab cells. The researchers themselves said it does not show any benefit from sniffing farts, and the gas is harmful in high doses.

MYTH 2 **"Fart gas is mostly methane."** Hydrogen, nitrogen and CO2 dominate. Methane showed up in only 3 of the 10 volunteers.

MYTH 3 **"Science says you fart 14 times a day."** That number is not in the study. The median was 8 daytime episodes, in 10 healthy volunteers.

MYTH 4 **"Farts travel at 10 feet a second."** A zombie fact: even HowStuffWorks calls it anecdotal, and a hobbyist's measurements came out around a thousand times slower. Treat it as unverified.

MYTH 5 **"People set their farts on fire and end up in hospital, says science."** The documented burns come from bowel gas igniting during surgery, with a laser or electrocautery in the room. We found no solid journal case of the home version. Please do not try it.

BONUS MYTH

"The loudest fart has a world record." Guinness has no official category for farting, so online records are unofficial and conflict. Burps are verified: Neville of Australia beat a 109.9 dB record in 2021.

THE 300-POOL MATHS

8.2 billion people x 90 mL = about 738,000 m3. One Olympic pool is 2,500 m3, so about **295 pools**. That leans on a 90 mL median from 10 people, so say "about 300".

GAS TRIGGERS TRACKER

What I ate	Gas (0–3)	Notes
Baked beans (top gas maker for most people, per researcher Levitt)		

Holding it in? Levitt does not recommend it: the downsides are discomfort, bloating and belching, and eventually an uncontrolled release. Cows, meanwhile, release most of their methane as burps, not farts.



Sources Lead Stories (Exeter AP39 hoax debunk) · Tomlin et al., Gut 1991 · HowStuffWorks, flatology.com (speed) · Gizmodo AU, PMC3569892 (surgical ignition) · Salon 2000 and Discover (Levitt) · Guinness record coverage (97x.com)